

genetics



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E-I

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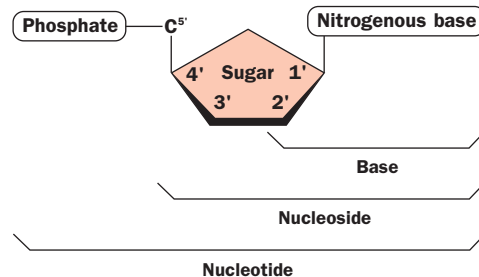
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For Your Reference

The following section provides a group of diagrams and illustrations applicable to many entries in this encyclopedia. The molecular structures of DNA and RNA are provided in detail in several different formats, to help the student understand the structures and visualize how these molecules combine and interact. The full set of human chromosomes are presented diagrammatically, each of which is shown with a representative few of the hundreds or thousands of genes it carries.

NUCLEOTIDE STRUCTURE



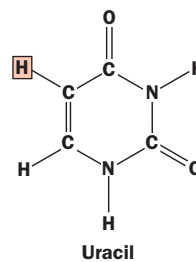
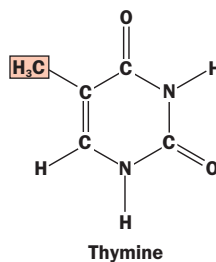
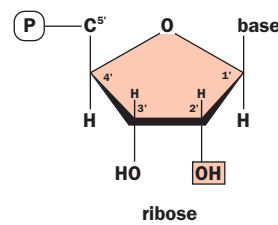
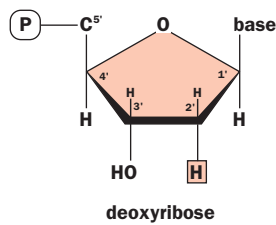
Sample naming conventions for each structure:

Adenine

Adenosine

Adenosine monophosphate

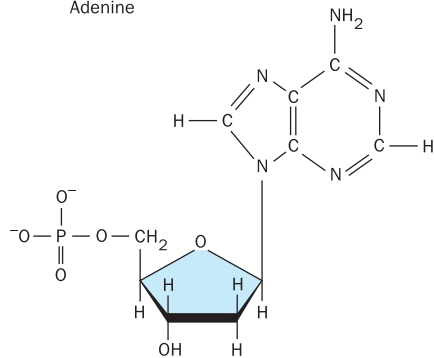
DNA VS. RNA



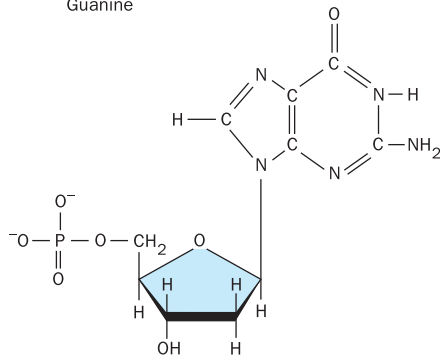
NUCLEOTIDE STRUCTURES

Purine-containing DNA nucleotides

Adenine

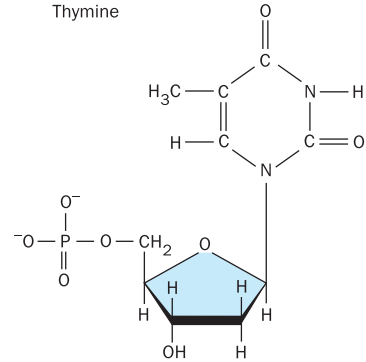


Guanine

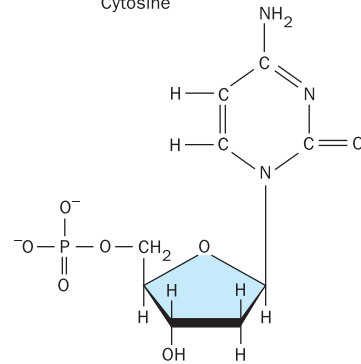


Pyrimidine-containing DNA nucleotides

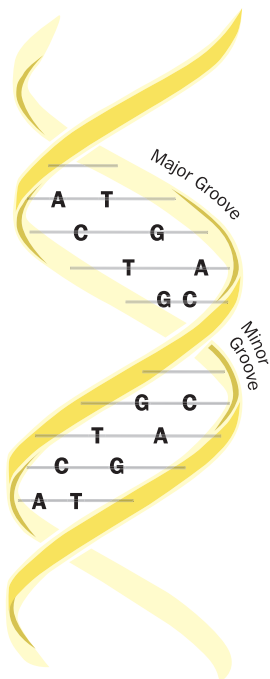
Thymine



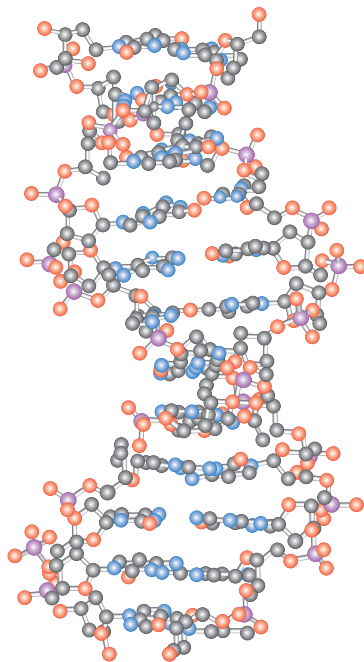
Cytosine



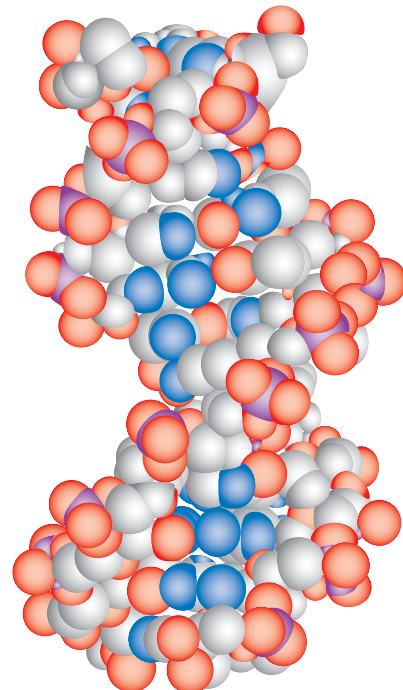
CANONICAL B-DNA DOUBLE HELIX



Ribbon model

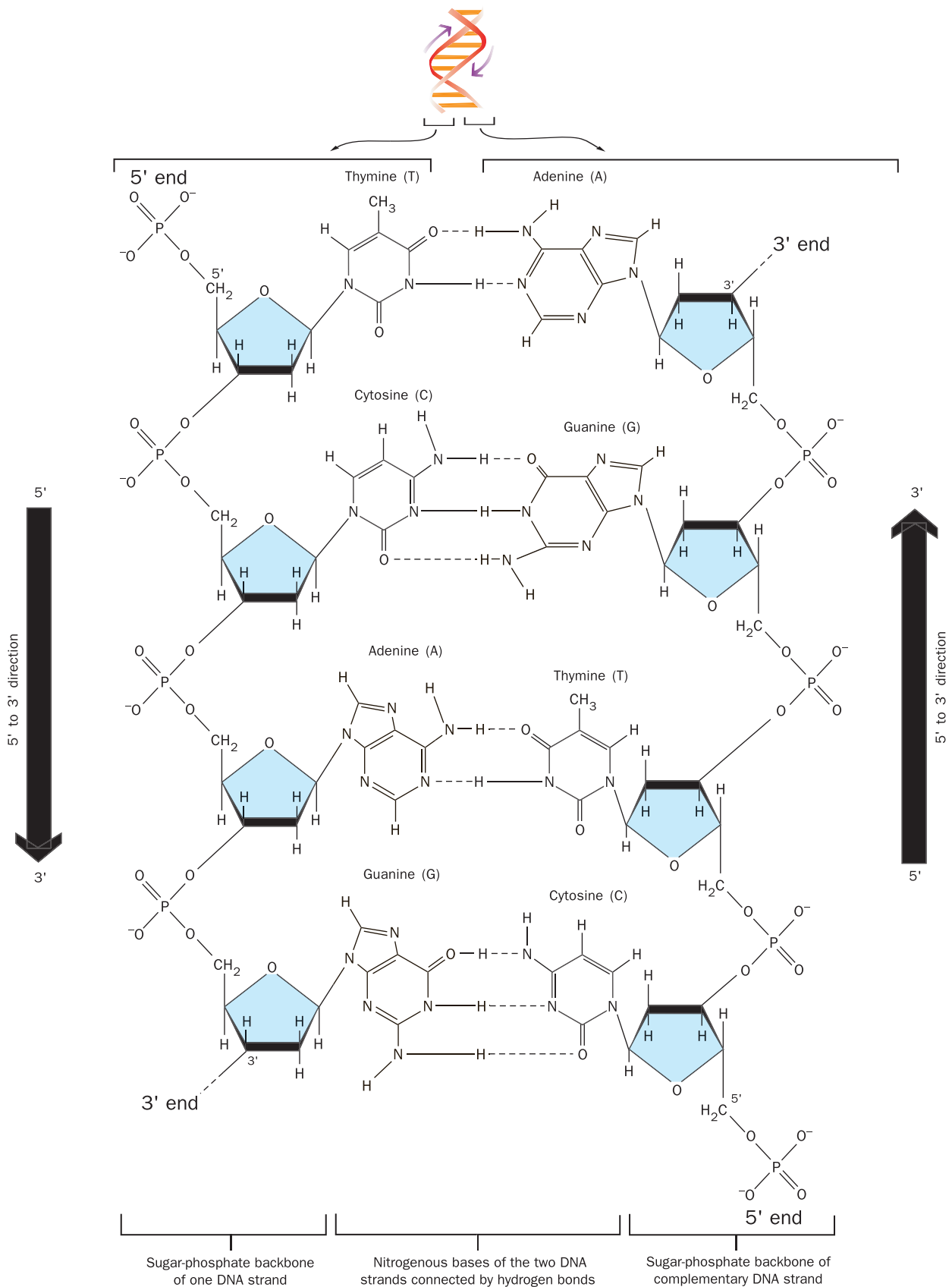


Ball-and-stick model

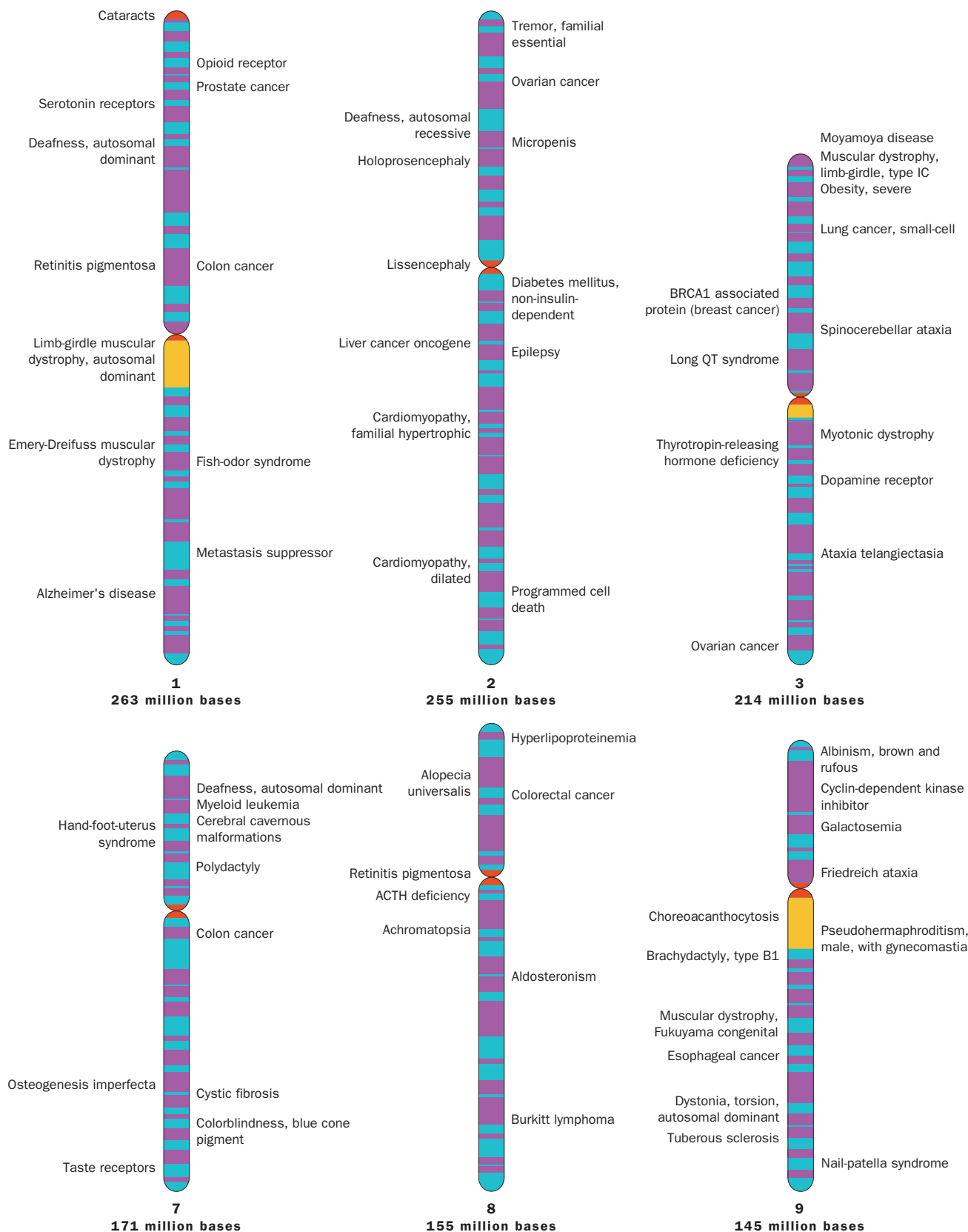


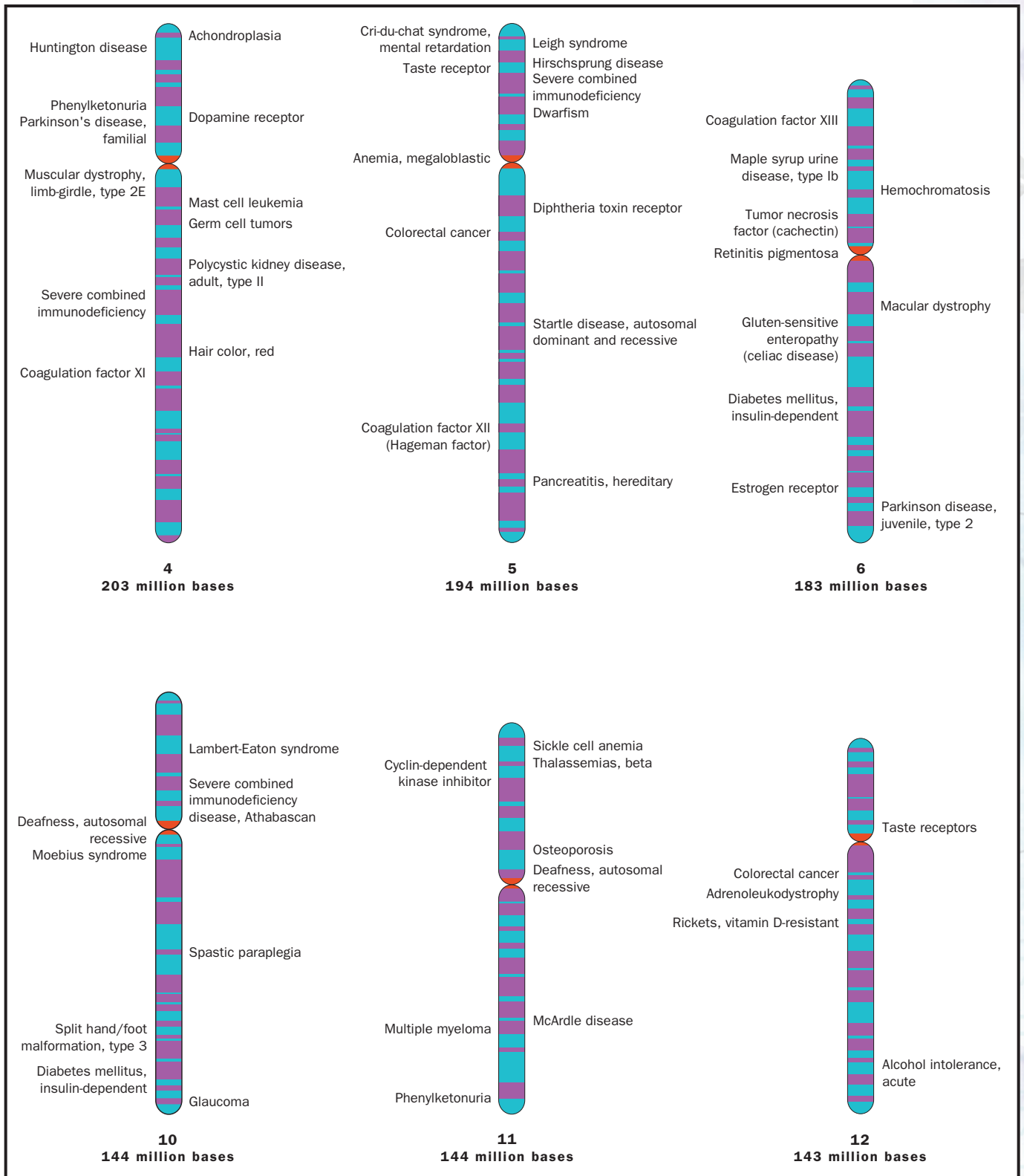
Space-filling model

DNA NUCLEOTIDES PAIR UP ACROSS THE DOUBLE HELIX; THE TWO STRANDS RUN ANTI-PARALLEL

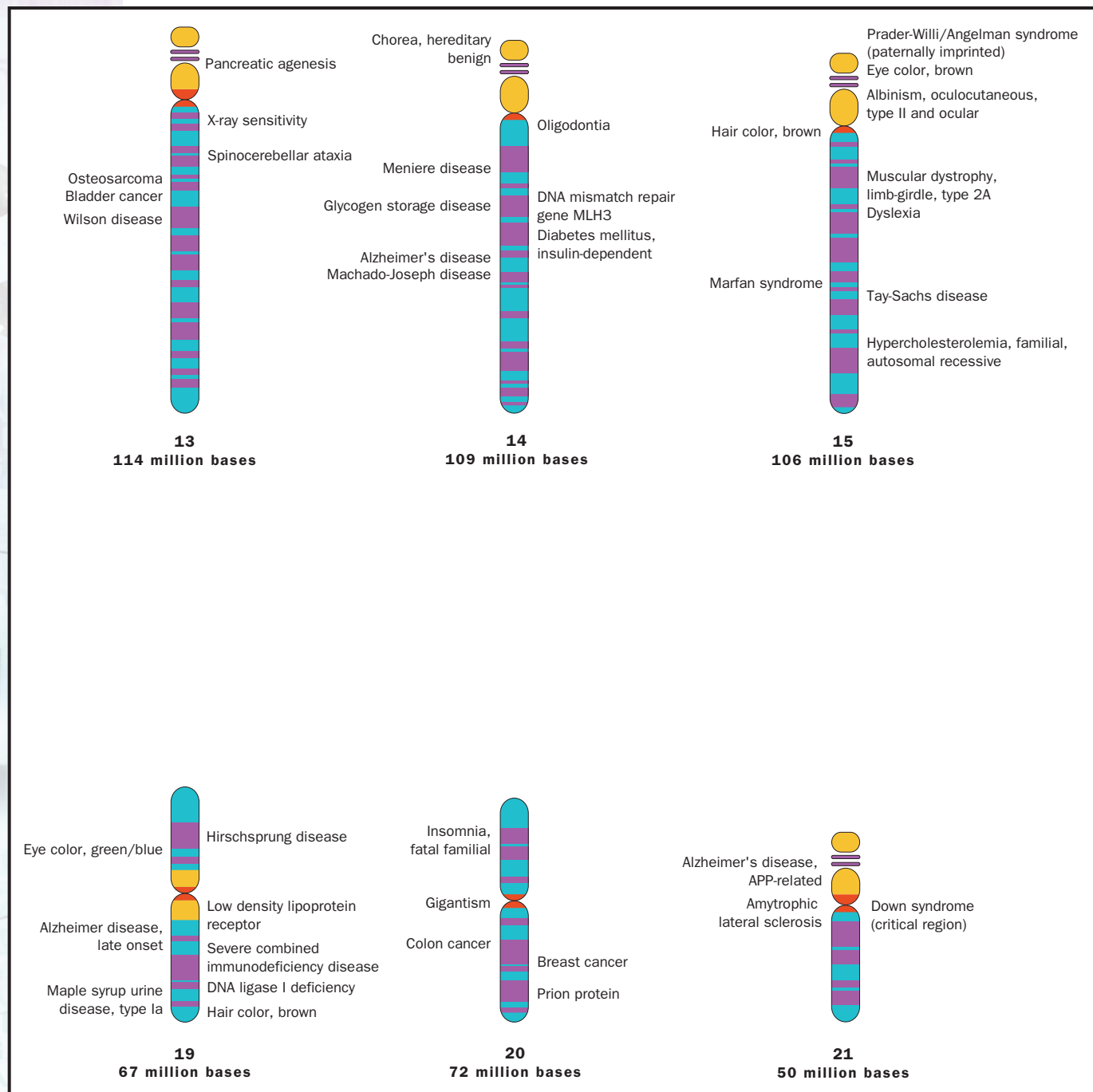


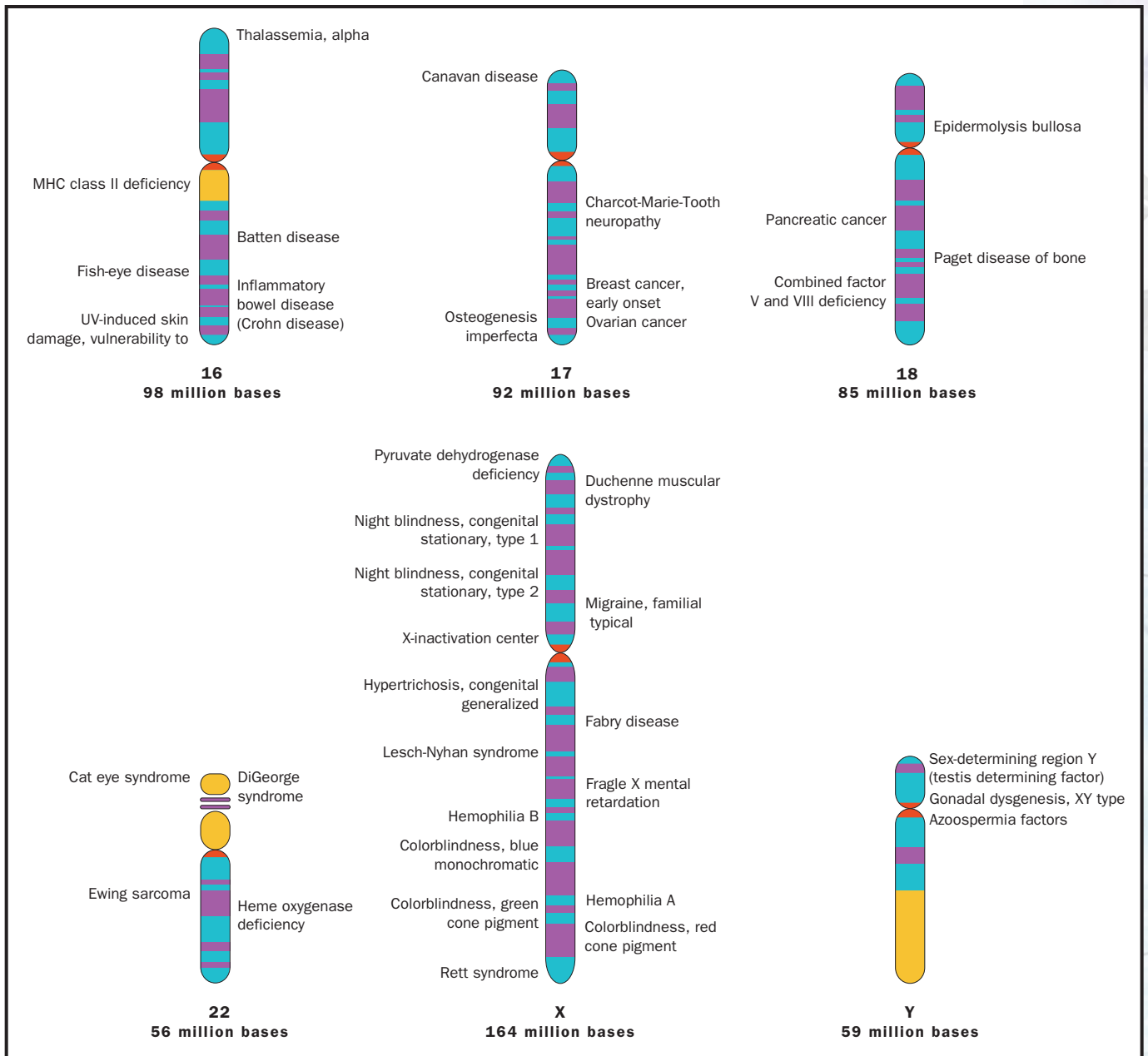
SELECTED LANDMARKS OF THE HUMAN GENOME





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Educator

An educator is a person who systematically works to improve another's understanding of a topic. The role of educator encompasses both those who teach in classrooms and the more informal educators who, for example, work in zoos, museums, and recreational areas. The work of educators varies depending on the institution that employs them and the age or grade level of the people he or she teaches.

K-12 Educator

Educators who teach children from kindergarten to 12th grade have many responsibilities. They must prepare lesson plans for all of their classes, tailoring their plans to the instructional backgrounds and abilities of the children in each class. Class sizes may range from twenty to thirty-five students, and a teacher may be assigned as many as six different classes. Thus, in addition to teaching, educators at this level must be skilled in classroom management, for they must often deal with behavioral problems and even act as surrogate parents for younger children.

Class preparation includes drafting lesson plans, writing and grading tests, evaluating homework, and sometimes purchasing classroom supplies with their own money. Teachers are held accountable for the success of their students, and are required to participate in professional development programs (courses or seminars) throughout their careers. Finally, they must maintain communication with the parents, participate in numerous educational and administrative activities, and sponsor student organizations such as the student council, sports teams, and school clubs.

Two-Year College Faculty

Teachers at two-year colleges are responsible for preparing lessons for two to four different classes, and they usually teach between four and five courses each semester. They, too, have mandatory professional development. In addition to their teaching duties they must serve on school committees dealing with such issues as hiring or technology. They are responsible to their department heads, deans, and other higher administrators, but they generally do not have to deal with parents.



Class sizes vary from twelve to thirty students, depending upon the subject and type of course being taught. Lecture courses, for example, tend to have higher enrollments than laboratory courses. For lab courses, educators sometimes may have to do their own set-up, if the department does not employ a laboratory technician. Some two-year schools hire their faculty with the possibility of tenure, whereas others offer only short-term contracts, typically for one to three years. Higher positions are in the administration.

Four-Year College Faculty

Educators at a four-year college must conduct publishable research, direct a laboratory, and mentor graduate students, in addition to teaching one to four undergraduate courses each semester or year. On first joining a faculty, educators typically receive the rank of assistant professor. On contract renewal, promotion to associate professor is the norm, with the rank of full professor being accorded upon achieving tenure.

Graduate student teaching assistants (TAs) are often available to assist the educator. TAs usually do all of the grading and supervise the lab classes. Four-year college educators also usually have the assistance of a laboratory technician.

To receive tenure, educators at this level must pass a review by a committee of their peers in their department, in which their success in research, publications, and teaching is evaluated. Most departments require their faculty to serve on one or more internal committees, and service on college- or university-wide committees may also be required, particularly of junior (non-tenured) faculty.

Educational Requirements

Educators at the K-12 level must have, minimally, a bachelor's degree in education. Often, however, they earn a degree in an academic subject and take master's level courses in education. Public school educators must pass a state-administered test, after which they receive certification allowing them to teach. Private schools may not require certification.

Educators at two- and four-year colleges usually have degrees in their area of specialization, but state certification is not required. Some two-year institutions may require master's or doctoral training in education in addition to a degree in an academic subject, but educators at the college level usually gain their teaching experience as teaching assistants. In some cases, an academic degree is not required to teach in a two-program, if the teacher has had enough experience working in the field for which students are to be trained. Four-year colleges expect their educators to hold a graduate degree, often a doctorate, and usually require a record of published research.

Compensation and Other Benefits

Pay scales for educators vary depending upon the grade level taught and the institution at which they teach. At the K-12 level, teachers can earn from \$21,000 to \$80,000, depending on years of experience and the school district in which they teach. At two-year colleges, starting salaries average about \$35,000, but the upper salary limit is about the same as for K-12. Four-year colleges and universities typically offer starting salaries of about \$40,000,

and can rise to more than \$100,000 for full professors whose research and publications have earned them renown in their fields.

Educators typically are free of teaching duties during the summer months, but those who work at the college level are expected to conduct research and publish even when classes are not in session. College-level teachers are also often eligible for sabbaticals, during which teaching duties are not required, but they are expected to use this time for a research project. A love of teaching and learning is needed for educators at all levels, but each student age group makes different demands on the teacher. **SEE ALSO** COLLEGE PROFESSOR.

Linnea Fletcher

Embryonic Stem Cells

Mammalian embryonic stem (ES) cells have the special property of being able to differentiate into virtually every cell type. Because ES cells can be genetically manipulated **in vitro** and can be transplanted into embryos and adults, they are a powerful tool in biological experiments and hold promise for future medical therapies.

The ability to differentiate into all cell types, a property known as pluripotency, arises from the fact that ES cells are isolated from in vitro outgrowths of early stage embryos (in the mouse, at three and one-half days, at the **blastocyst** stage). These outgrowths are cultured in specialized conditions—often in the presence of support cells, called feeder cells, which do not proliferate, and specific growth factors. The ES cells proliferate rapidly in culture, and clonal (identical) populations can readily be initiated from single cells.

Until 1998 the only mammalian embryonic stem cells isolated were those from the mouse. The first human embryonic stem cells were isolated in 1998 from embryos created through in vitro fertilization. In the United States, government funding for research on human ES cells was restricted in 2002 to a small set of existing human ES cell lines. Only privately funded research remained able to use other sources of cells.

In the following years, most publicized research on human ES cells will concentrate on manipulating cells in culture, attempting to understand the signals that cells require to proliferate and differentiate into various adult cell types, such as nerve or muscle cells.

In 2001 there was a highly publicized report of human cloning using human ES cells, but the resulting embryos did not progress past the twelve-cell stage, and there was no reliable evidence that the ES cells ever contributed to the development of the embryos. Nonetheless, most researchers thought obstacles would likely be overcome, and that the use of ES cells for therapy or cloning would become technically possible.

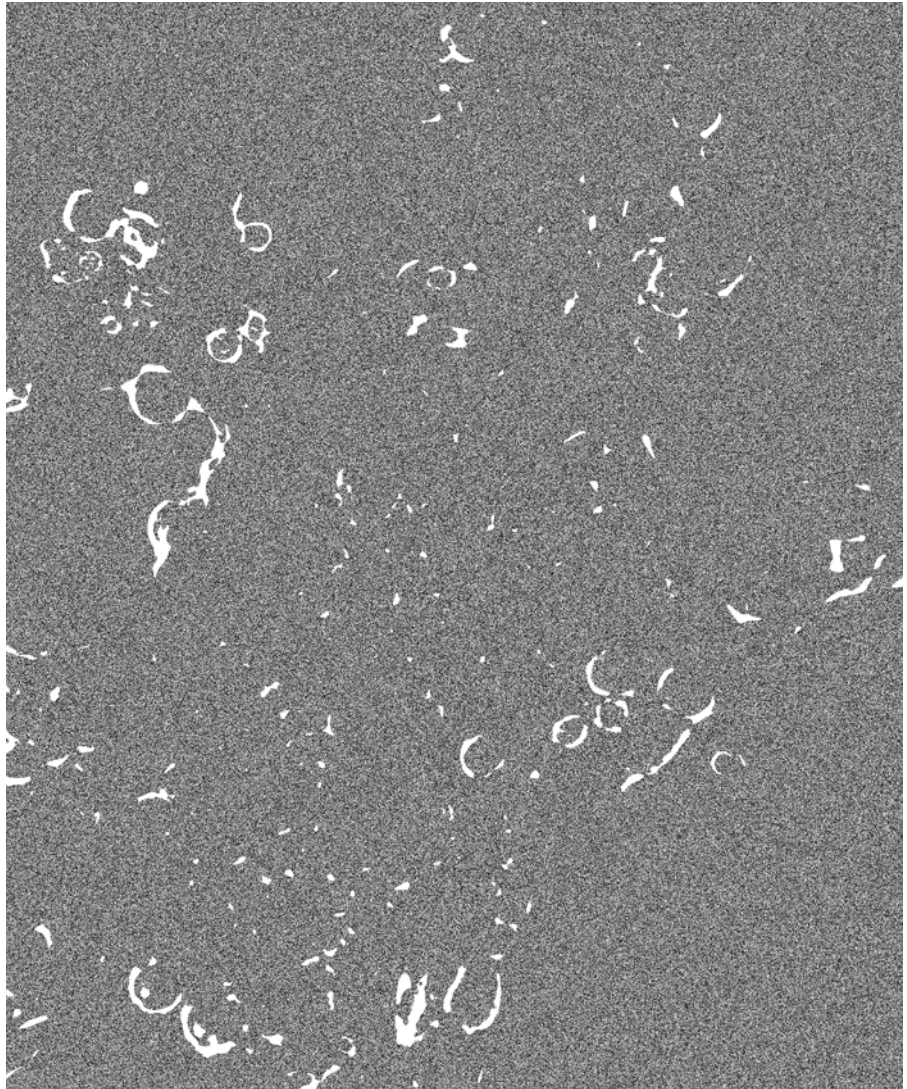
Mouse ES cell research is much further advanced. Research in the mouse has included injection of ES cells into blastocysts, an early stage in embryo development. Once injected, ES cells fully participate in embryonic development and form part of every tissue in the embryo. The resulting mouse is called a chimera, since it usually contains a mixture of cells derived from

in vitro “in glass”; in lab apparatus, rather than within a living organism

blastocyst early stage of embryonic development



Figure 1. Mouse embryonic stem cells in culture.



both the ES cells and the host blastocyst. An important feature of a chimera is that its ES cells will contribute to forming the germ cells, which eventually form the gametes of the animals. Consequently, mating of the chimera with recipient females allows the (ES cell derived) sperm to deliver their genetic material into an egg, and subsequent offspring will carry genes from the ES cell. Thus genetic manipulations made in ES cells in culture can be transmitted through chimeras to intact animals.

As with other cell lines grown in vitro, it is possible to add new genes, termed “transgenes,” or to modify existing genes in ES cells using a technique known as gene targeting. Gene targeting involves using specialized DNA **vectors** that share substantial DNA sequence similarity with the gene that needs to be modified. When introduced into cultured ES cells, the gene-targeting vector undergoes “homologous recombination,” a process in which the vector is integrated into the existing chromosomal gene and leaves the ES cells genetically modified.

Removing a gene product is often referred to as creating a “knockout,” since it prevents a protein from being expressed. Modifying a specific gene

vectors carriers



Figure 2. A typical chimera. ES cell-derived tissue is identified as light patches in the mouse coat.

in the genome to include a single DNA nucleotide change (mutation) or incorporating a genetic marker (such as a gene that produces a colored protein that can be seen with a microscope) is accomplished by a similar method and is often referred to as creating a “knock-in”—since a piece of DNA is inserted into a specific part of the genome.

The genetically modified ES cells can be used to create mouse chimeras, whose offspring will carry the genetic modifications into the ES cells. These mice can be identified by analysis of DNA made from a small tissue sample, usually taken from the mouse’s tail. The genomic DNA can be analyzed by polymerase chain reaction (**PCR**) or **Southern blot** to identify the mice carrying the genetic change. These offspring will be **heterozygous** for the introduced mutation and can be interbred to generate a second generation, some of which will be **homozygous** for the introduced mutation. The homozygous mice will lack both copies of the normal gene—any differences to normal development displayed by the mice as a result of losing the gene of interest can be analyzed. This, in turn, can provide valuable clues to the function of the gene and its normal role in mouse physiology.

This experimental approach to genetic manipulation of mice, which was pioneered in the 1980s, has been used to modify thousands of mouse genes and has played a significant role in understanding many physiological and pathological processes. For example, mutations in the p53 gene show its involvement in cancer, and mutations in the psd-95 gene show involvement in learning and memory. Variations on these methods are also used to model human disease mutations in mice and to test drug therapies.

ES cells also have potential for treating human diseases through “cell-based” therapy. In this context, the ES cells (or their differentiated derivative cells) are transplanted into patients whose own cells are defective or degenerated. This strategy could have applications for diseases such as Parkinson’s disease (which affects brain cells), diabetes (pancreatic cells) and heart disease (which affects heart muscle cells). An important clinical issue

PCR polymerase chain reaction, used to amplify DNA

Southern blot a technique for separating DNA fragments by electrophoresis and then identifying a target fragment with a DNA probe

heterozygous characterized by possession of two different forms (alleles) of a particular gene

homozygous containing two identical copies of a particular gene

at this point will be whether ES cells not derived from the patient will be rejected by the patient's immune system. If so, one strategy for dealing with this problem would be to use a patient's own cells to create an embryo by nuclear transfer, from which ES cells compatible with that patient could then be derived.

The use of human blastocysts in both research and therapy remains controversial because it is necessary to destroy the blastocyst to generate an ES cell line from it. As a result, work with human embryos is governed by strict regulations in many countries. SEE ALSO GENE TARGETING; MARKER SYSTEMS; REPRODUCTIVE TECHNOLOGY; RODENT MODELS; TRANSGENIC ANIMALS.

Seth G. N. Grant and Douglas J. C. Strathdee

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Epidemiologist

Epidemiologists are scientists that study the factors influencing the health status of populations. These populations may be defined by geography (such as the residents of a particular city), occupation (such as members of the armed forces), or any other common trait (such as age, race, or sex). Epidemiologists look for trends in measures of the health of the population, such as the average life span, the leading causes of death, and the number of cases of a disease that are found in the population. To determine what causes certain trends or health problems, epidemiologists collect large amounts of data about individuals in the population. They analyze these data to determine who is sick, when they got sick, and what factors the sick people have in common. The process is similar to that which investigators use to search for clues to solve a crime. Thus, epidemiologists are often described as "disease detectives."

Career Requirements, Employment, and Compensation

A career in epidemiology generally requires a master's or doctoral degree in public health. Epidemiologists work in many different settings: universi-